Yanjun Sheng

PhD Candidate in Astronomy & Astrophysics
Research School of Astronomy & Astrophysics, ANU
Yanjun.Sheng@anu.edu.au | github.com/Yanjun-Sheng
ORCID: 0009-0004-1470-5201 | ADS: ADS Library

Education

PhD in Astronomy & Astrophysics — Aug 2023 – Present
Australian National University, Canberra, Australia — Supervisor: Yuan-Sen Ting

- Thesis (working title): *"Constraining the properties of the Milky Way's halo with the Large Magellanic Cloud"*
- Expected completion: May 2027

MSc in Astronomy & Astrophysics — Jul 2021 – Jun 2023
Australian National University, Canberra, Australia — Supervisor: Yuan-Sen Ting

- Thesis: *"Unraveling the first-infall history of the Large Magellanic Cloud using N-body simulations"*

BSc in Astronomy — Sep 2017 – Jun 2021
Beijing Normal University, Beijing, China

Research Interests

Milky Way–LMC interactions; stellar-halo dynamics and tidal streams; N-body simulations and galaxy modeling; simulation-based inference; Galactic archaeology, near field cosmology and stellar spectroscopy.

Publications

Published / Accepted

1. **Sheng, Y.**, Ting, Y.-S., Różański, T., Xue, X.-X., Crocker, R. M., & Pan, J. (2026). *LMC-induced Perturbations in the Milky Way Halo II: Bridging Field-level Inference and Summary-level Simulation-Based Inference*, accepted for publication in *MNRAS*. arXiv:2607.20006
2. **Sheng, Y.**, Ting, Y.-S., & Xue, X.-X. (2025). *LMC-induced Perturbations in the Milky Way Halo: I. HaloDance Simulation Suite and Observational Forecasts*, *MNRAS*. arXiv:2507.03663
3. **Sheng, Y.**, Ting, Y.-S., Xue, X.-X., Chang, J., & Tian, H. (2024). *Uncovering the first-infall history of the LMC through its dynamical impact in the Milky Way halo*, *MNRAS*. arXiv:2404.08975

In Preparation

4. **Sheng, Y.** et al., "LMC-induced Perturbations in the Milky Way Halo III: Measuring the Milky Way–LMC System with H3 and SDSS-V", in preparation.

Research Experience

PhD Researcher — Aug 2023 – Present
Australian National University, RSAA — Canberra, Australia

- Built and ran the HaloDance suite of ∼3,000 high-resolution N-body simulations to map the dynamical response of the Milky Way stellar halo to the LMC.
- Designed robust statistical summaries of velocity moments, anisotropy, and spatial structure, enabling direct comparisons between simulations and stellar-halo observations.
- Developed a production simulation-based inference pipeline combining field-level and summary-level approaches to constrain Milky Way–LMC parameters from phase-space data.
- Currently applying the inference pipeline to H3 and SDSS-V stellar surveys to obtain observational measurements of the Milky Way–LMC system.

Observing Experience — 2023 – 2024
KMTNet, Siding Spring Observatory — 80 nights

- Conducted target acquisition, telescope focusing, and real-time data-quality assessment during 80 nights of observing.

Selected Talks

- "Constraining the Mass Distribution of the Milky Way and the Large Magellanic Cloud from the Phase-Space Structure of the Stellar Halo", *How Stars Move Around and What They Tell Us About Galaxies*, Sexten Center for Astrophysics, Sesto (Sexten), Italy, Jul 2026 (contributed talk).
- "Unveiling Milky Way-LMC Properties from All-Sky Kinematics of the Galactic Halo", *SDSS-V Collaboration Meeting*, Heidelberg, Germany, Jun 2025 (contributed talk).
- "Unveiling Milky Way-LMC Properties from All-Sky Kinematics of the Galactic Halo", *XMC II: Clouds over Yellowstone*, Bozeman, USA, May 2025 (contributed talk).

Grants, Scholarships & Awards

- RSAA HDR Travel Fund
- RSAA PhD Scholarship (2023–2027), RSAA/ANU
- RSAA Supplementary Scholarship (2023–2027), RSAA/ANU
- RSAA Masters (Advanced) Scholarship (2021–2023), RSAA/ANU

Teaching & Service

- Tutor for Stars (ASTR3007), Semester 1, 2026.
- Referee for *The Astrophysical Journal* (2025).

Technical Skills

Programming	Python, Bash, C, SQL
Scientific tools	NumPy, SciPy, Astropy, pandas, h5py
ML / Inference	PyTorch, simulation-based inference, normalizing flows, flow matching, neural networks, MCMC
Simulations	GADGET-4, GalIC, N-body methods, initial-condition generation, halo potentials

References

Yuan-Sen Ting	Associate Professor, The Ohio State University	ting.yuansen.astro@gmail.com
Roland Crocker	Associate Professor, Australian National University	roland.crocker@anu.edu.au
Xiang-Xiang Xue	Professor, National Astronomical Observatories, Chinese Academy of Sciences	xuexx@nao.cas.cn